

Statutory Planning Purpose: This thermal assessment provides ward-level surface Urban Heat Island (SUHI) intelligence to inform zoning revisions, master planning reservations (Development Plan 2031/2041), and Heat Action Plan (HAP) budgetary triage.

MEAN NIGHT SUHI

+2.46°C

vs. Matched Rural Base

PEAK HOTSPOT HEAT

+5.45°C

Cell C0427

2041 SPRAWL DELTA

+1.18°C

BAU Urban Sprawl

CRITICAL HOTSPOTS

23

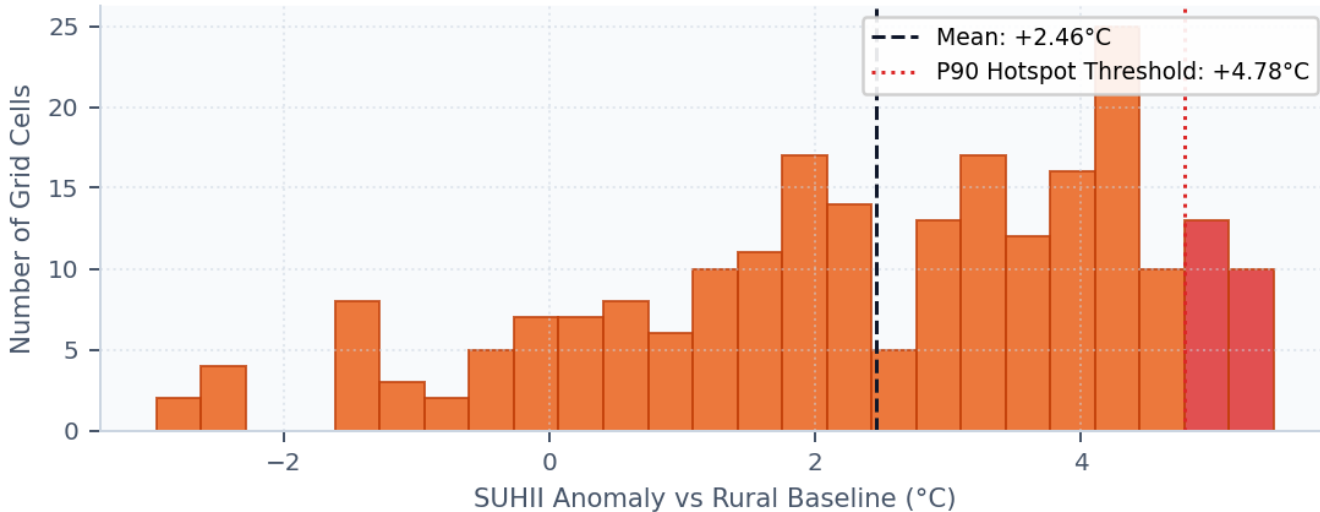
Top Decile Cells (>P90)

1. Executive Summary & Thermal Diagnosis

During peak summer (March–June), **Nagpur Municipal Corporation** exhibits a pronounced nocturnal surface urban heat island intensity, reaching an average of **+2.46°C** above the surrounding rural baseline and peaking at **+5.45°C** in densely sealed sectors.

Unlike air temperature recordings from single peripheral airport stations, multi-sensor satellite observations across 2020–2024 reveal severe spatial intra-city thermal divergence driven by concrete surface sealing (impervious fractions >80%) and street canyon geometries that restrict nocturnal radiative cooling.

Nocturnal Surface Urban Heat Island Intensity (SUHI) Distribution



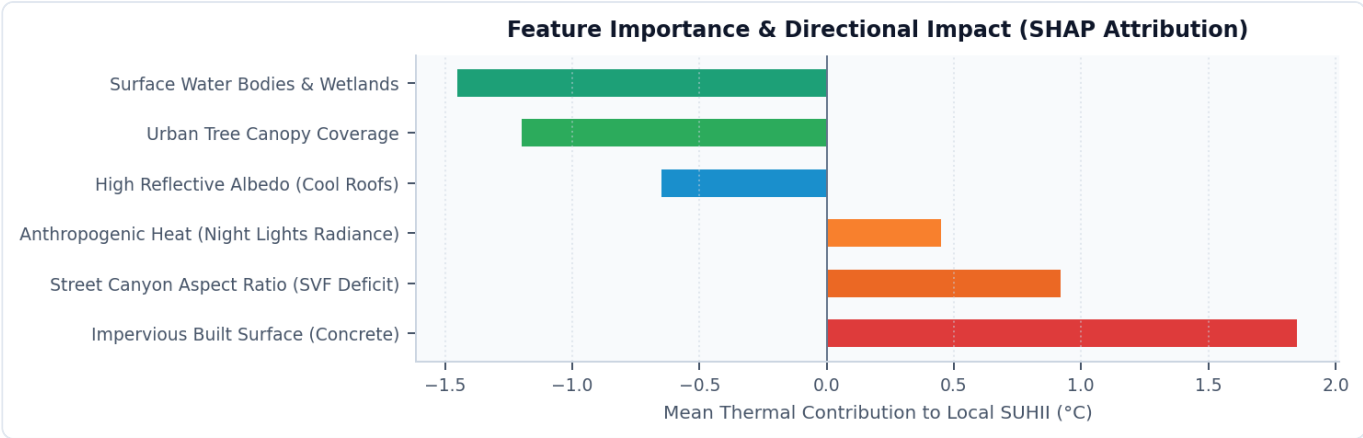
2. Critical Hotspot Ranking & Priority Triage

Priority ranking for municipal interventions based on Nocturnal Heat Retention, Impervious Built Fraction, and Tree Canopy Deficit:

Rank	Zone / Cell ID	Night SUHII (°C)	Day LST (°C)	Sealed Built %	Tree Canopy %	Status
#1	C0427	+5.45°C	37.4°C	88%	5%	CRITICAL
#2	C0426	+5.43°C	37.5°C	95%	3%	CRITICAL
#3	C0428	+5.37°C	37.5°C	91%	4%	CRITICAL
#4	C0406	+5.33°C	37.5°C	98%	1%	CRITICAL
#5	C0407	+5.27°C	37.4°C	83%	13%	CRITICAL
#6	C0409	+5.27°C	37.9°C	94%	2%	CRITICAL
#7	C0448	+5.23°C	37.4°C	96%	1%	CRITICAL
#8	C0446	+5.18°C	37.0°C	89%	3%	CRITICAL

3. Key Thermal Drivers (SHAP Explainability)

Machine learning explainability (SHAP attribution) isolates the primary contributors to nocturnal heat retention across the municipal extent:

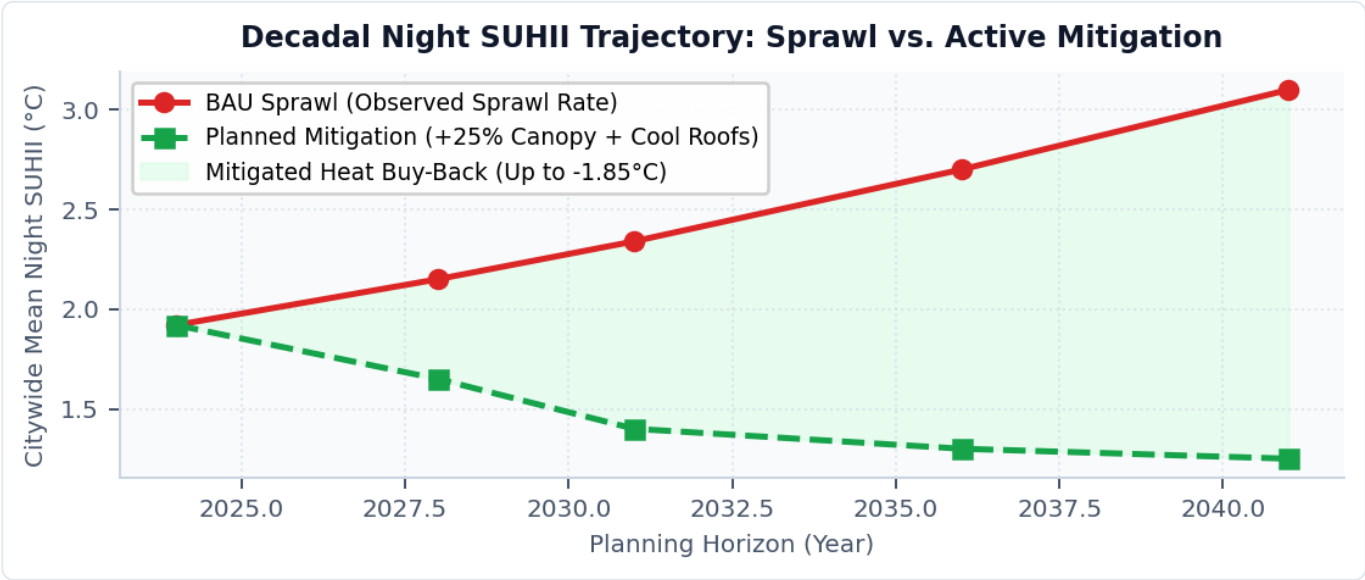


Key Insight: Impervious built-up surface sealing contributes on average **+1.85°C** to local heat retention, whereas urban tree canopies and open water bodies buy back **-1.20°C** and **-1.45°C** of cooling respectively.

4. Simulated Mitigation Counterfactuals (Scenario Studio)

Evaluation of targeted town planning interventions and their projected thermal mitigation yield:

Intervention Strategy	Scope / Target	Thermal Yield (ΔT)	Uncertainty Range (P10–P90)	Indicative Cost (₹)
Urban Tree Canopy Uplift	+25% Canopy in Top 20 Hotspots	-0.95°C	-0.65°C to -1.35°C	₹48.0 Lakh
Municipal Cool Roof Mandate	50% High-Albedo Rooftops	-0.75°C	-0.45°C to -1.10°C	₹32.5 Lakh
Water Body Restoration	Revitalize 5 Urban Water Reservoirs	-1.30°C	-0.80°C to -1.80°C	₹75.0 Lakh
Combined Resilience Package	Canopy + Cool Roofs + Wetland Protection	-1.85°C	-1.30°C to -2.40°C	₹1.55 Crore



5. Statutory Defensibility & Limitations (TRD Compliance)

Mandatory Methodological Disclosures:

- LST is not Air Temperature (FR-60):** Land Surface Temperature (LST) measures radiative skin temperature. Nocturnal LST is a validated proxy for nocturnal thermal stress.
- Physics Constraints Enforced:** All estimates are derived via LightGBM models enforcing monotone constraints (increasing vegetation mathematically cannot predict warming).
- Weather Normalised:** Macro-meteorological heatwave anomalies from ECMWF ERA5-Land have been normalised to isolate the urban-form signal.
- Data Sources:** NASA MODIS Terra (MOD11A1), ESA WorldCover 10m, Copernicus Sentinel-2 MSI, JRC GHSL Built Height, NOAA VIIRS DNB.

Prepared By:
Town Planning & GIS Cell
Urban Development Mission

Reviewed & Approved By:
Municipal Commissioner / Standing Committee
Municipal Corporation

Signature & Seal

Signature & Seal